

Program : Diploma in Tool & Die Engineering	
Course Code : 4117	Course Title: Computer Aided Moulds & Dies Drawing
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To familiarize students with detailed part drawings and assembly drawings of various types of Moulds& Die casting dies from the given component drawing with the aid of design principles learned from Moulds& Dies technology.
- To provide skills of 3D modeling using Auto CAD/PTC Creo/solidworks/NX Unigraphics/CATIA/Delcam software etc.

Course Prerequisites:

Topic	Course code	Course name	Semester
Design concepts of Moulds& Dies		Moulds & Dies Technology	3
Limits, Fits, G&D Tolerance, Surface Finish symbols and values		Tool Engineering drawing	3
Heat treatment processes & various tool steels		Material science & Metrology	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Prepare the part & assembly drawings of Hand injection moulds, compression moulds, transfer moulds& blow moulds.	9	Applying
CO2	Prepare detailed drawings of Two plate moulds.	12	Applying

CO 3	Demonstrate the design concepts and detailed drawings of advanced moulding techniques.	12	Applying
CO 4	Prepare detailed drawings of Die casting dies.	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3		3	3			3
CO3	3		3	3			3
CO4	3			3			3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Prepare the part & assembly drawings of Hand injection molding, compression molding, transfer molding & blow molding.		
M1.01	Apply skills in drafting and modeling in any of the above two software.	9	Applying
M1.02	Sketch a single cavity hand injection mold – Part drawings & Assembly		Applying
M1.03	Sketch compression molds showing horizontal & vertical flash		Applying
M1.04	Sketch Plunger & Pot transfer Molds.		Applying
M1.05	Sketch a simple blow mold		Applying
	3D Modeling and drafting shall be done by using any of the above-mentioned software.		

CO2	Prepare detailed drawings of two plate molds.		
M2.01	Design injection mould base.	3	Applying
M2.02	Design 2 plate injection mould		Applying
M2.03	Design injection mould with sleeve ejection & submarine gate.	3	Applying
M2.04	Design injection mould with stripper plate ejection.		Applying
M2.05	Design mould with side core	3	Applying
M2.06	Design moulds having threads. – Collapsible core/automatic rack & pinion design.	3	Applying
	3D Modeling and drafting shall be done by using any of the above-mentioned software.		
	Lab Exam – I	1.5	
CO3	Demonstrate the design concepts and detailed drawings of advanced moulding techniques.		
M3.01	Design split cavity injection mould with sleeve ejection	3	Applying
M3.02	Design split cavity injection mould with dog leg cam.	3	Applying
M3.03	Design mould with split cores - (angular acting)	3	Applying
M3.04	Design concept of Three plate molds with multi-impressions.	3	Applying
	3D Modeling and drafting shall be done by using any of the above-mentioned software.		
CO4	Prepare detailed drawings of die casting dies.		
M4.01	Design a Die Casting Die for a cold chamber system.	9	Applying
M4.02	Design a Die casting Die for a hot chamber system.		Applying
	3 D Modeling and drafting shall be done by using any of the above-mentioned software.		
	Lab Exam – II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	Pye.R.G.W., "Injection Mould Design", Affiliated East – west press pvt Ltd, 2000
R2	Briston and Gosselin, —Introduction to Plastics", Newnes - Butterworths, London, 1970
R3	Mills.N.J., —Plastics", ELBS, 1986
R4	Dominick V.Rosato and Donald V.Rosato., "Injection Moulding Hand Book", CBS Publishers & Distributors, Delhi, 1987 4
R5	Athalye.A.S., "Moulding of Plastics", Multi Tech Publishing Co., 1998

Online Resources:

Sl.No	Website Link
1	https://jasonwebb.io/2012/09/overview-of-mold-making-and-casting-techniques-and-materials/
2	https://www.sciencedirect.com/topics/materials-science/mold-design
3	https://www.3dsystems.com/quickparts/learning-center/injection-moldingbasics
4	https://www.custompartnet.com/wu/die-casting#tooling
5	http://www.cast-rite.com/engineering/die-casting-product-design